

African
Brain
Data
Network

Module 1 Introduction to Electrophysiology

Introduction to Neuroscience

23rd of July 2026 – Nicole Vissers

Why do we study the brain?

Neuroscience links biological mechanisms to behaviour.

And can answer questions like:

- How do we see?
- How do we learn?
- How do we move?
- How does the brain generate behaviour?



Also give more insights into disorders, injuries and development, such as:

Alzheimer's disease, Epilepsy, Mental Health, Stroke, Autism, ADHD

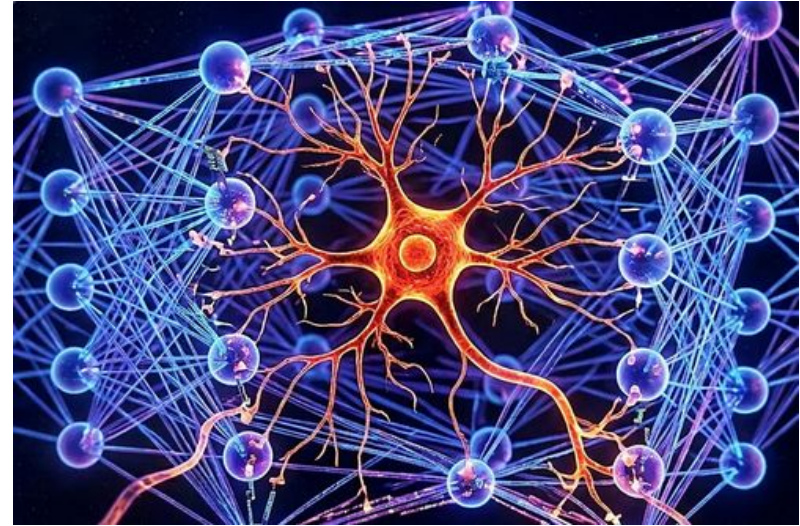
The brain: billions of interacting neurons

The brain consists of neurons

Human brain: ~86 billion neurons

Mouse brain: ~70 billion neurons

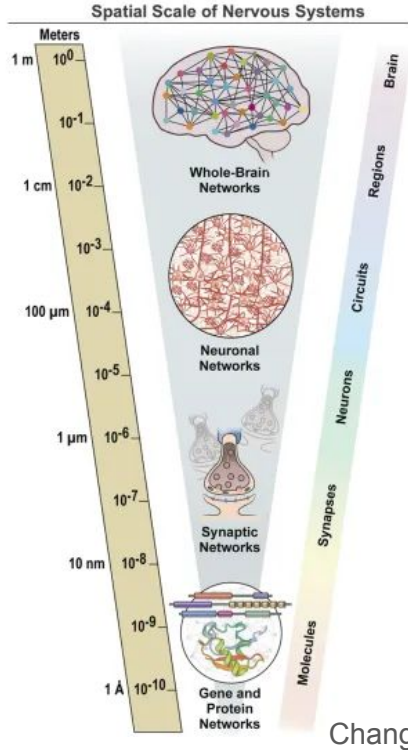
They are connected through synapses.



Christopher Lynn

Neuroscience can be done at several levels

A

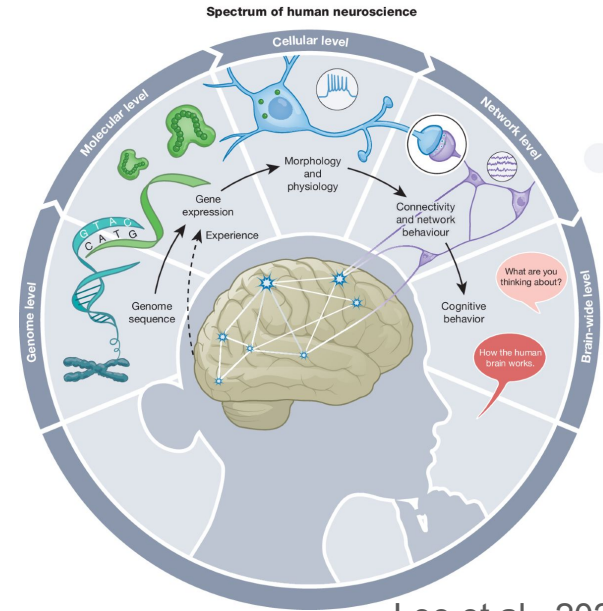
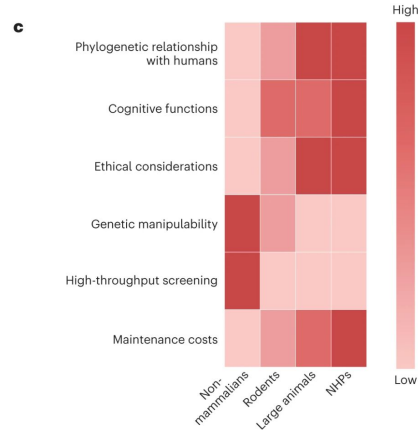
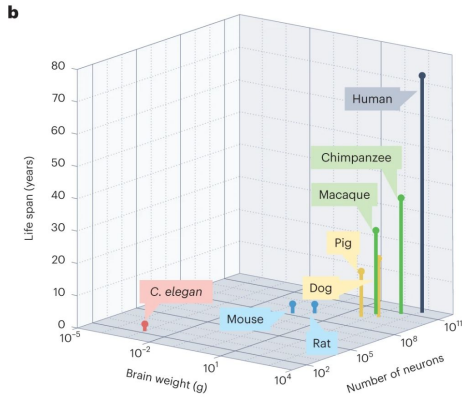
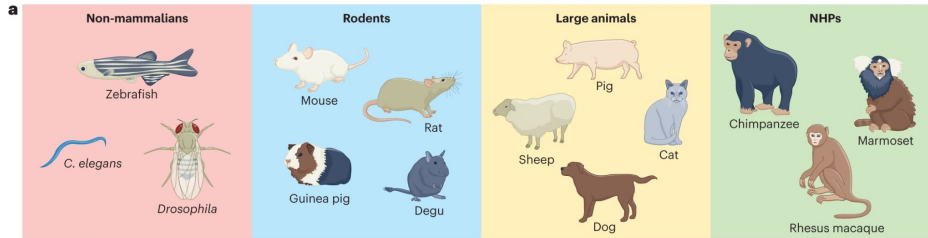


Chang, 2024

What level depends on the questions you are asking:

- Genes and proteins
- Synapses
- Single neurons
- Circuits
- Whole brain

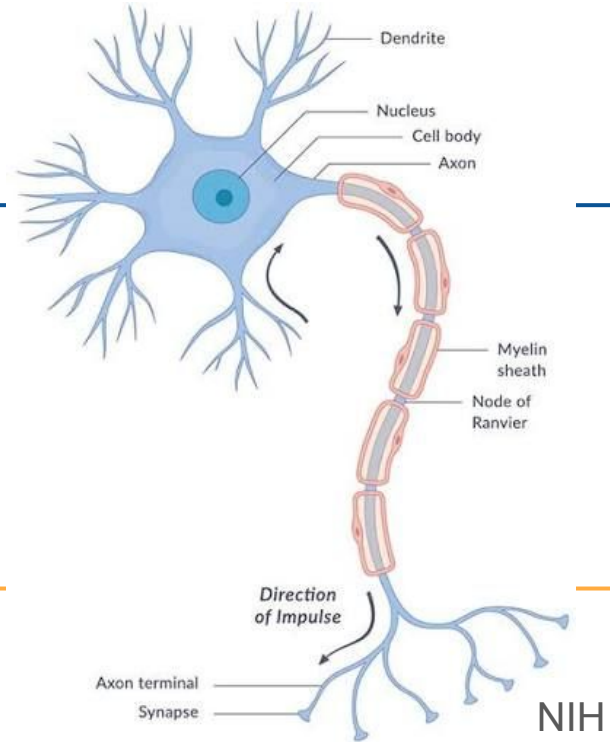
And can be studied in different organisms



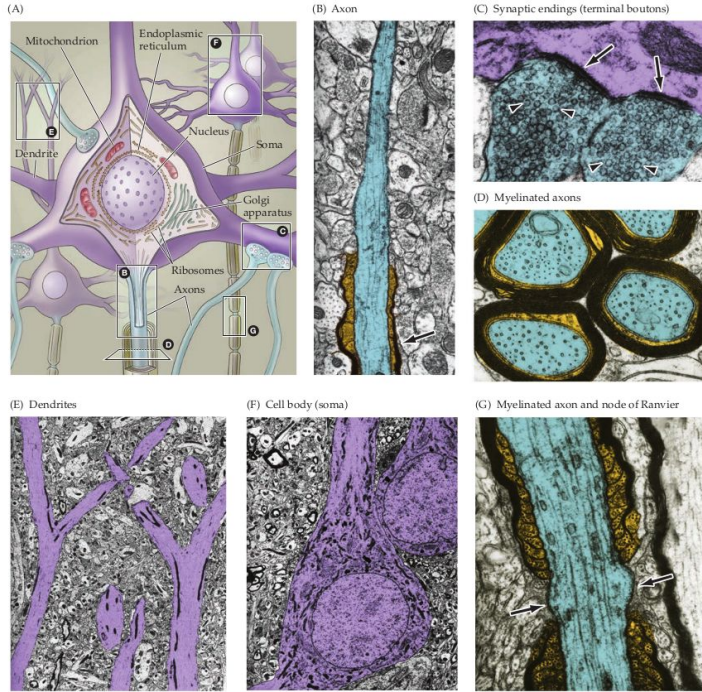
Lee et al., 2024

Padmanabhan & Gotz, 2023

The neuron



The neuron as a building block for the brain



A neuron consists of:

- Soma
- Dendrites
- Axon(s)
- Synapses

Purves, 6th edition

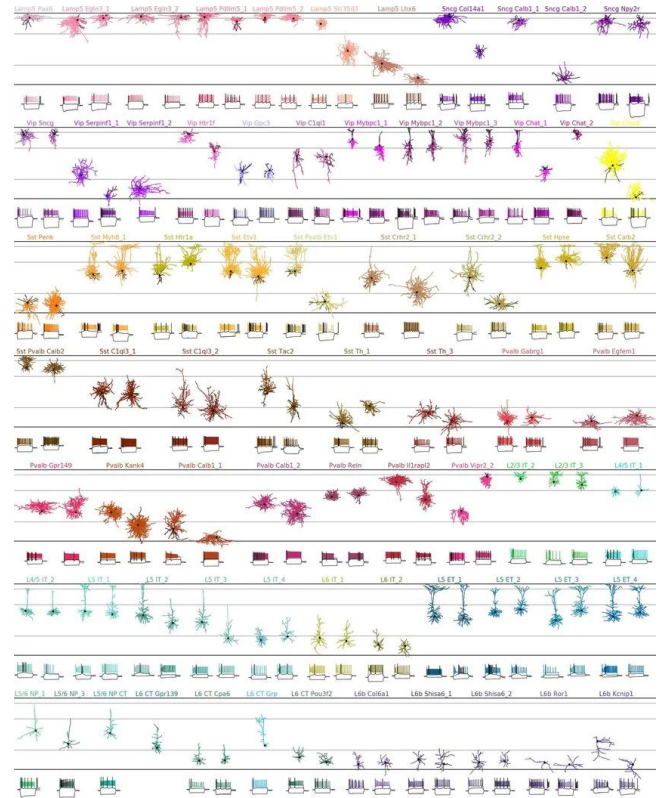
Different neurons have different jobs

Neurons with different roles:

- Sensory Neurons; detect outside world
- Motor neurons; control muscles
- Interneurons; connect neurons

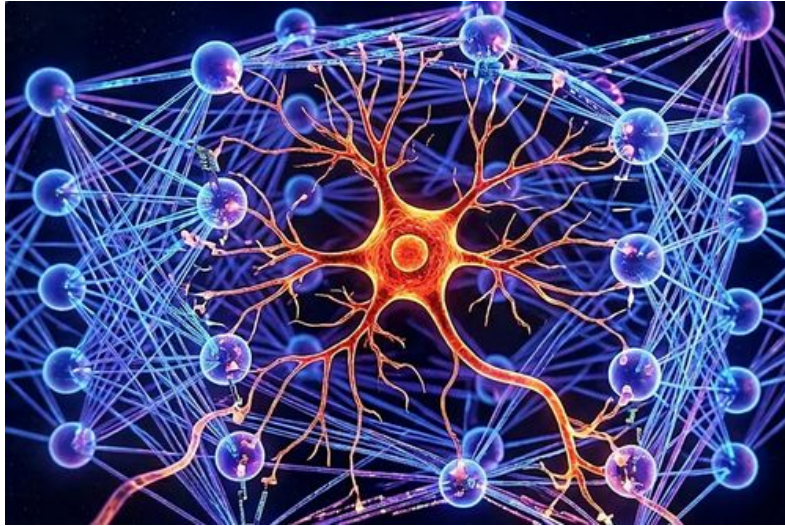
Neurons with different morphology;

→ Diversity of neurons in the motor cortex



Scala et al., 2021

Neurons form networks



Christopher Lynn

Datasets about connectivity:

Blue Brain Project;

<https://portal.bluebrain.epfl.ch/>

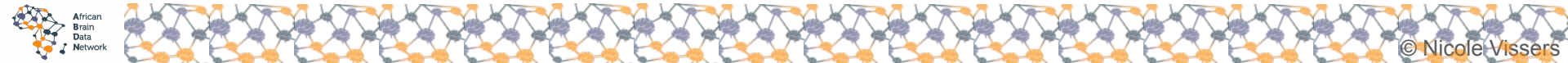
MICrONS;

<https://www.microns-explorer.org/>

Studying the **connectome**

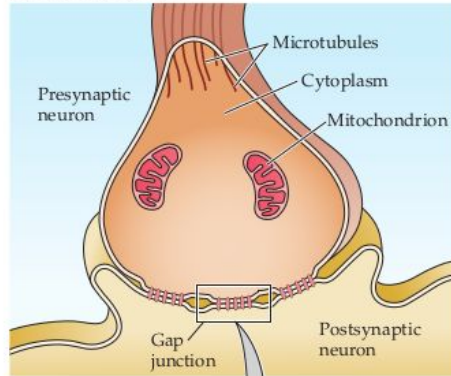


Electrical properties of neurons

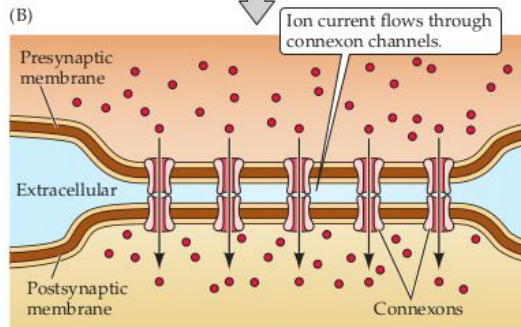


Neurons communicate through synapses

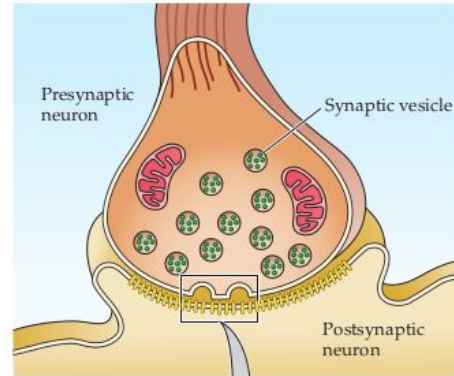
(A) Electrical synapse



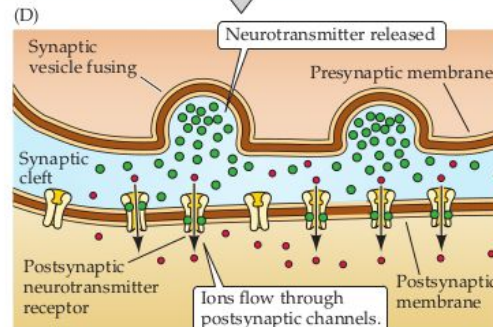
(B)



(C) Chemical synapse

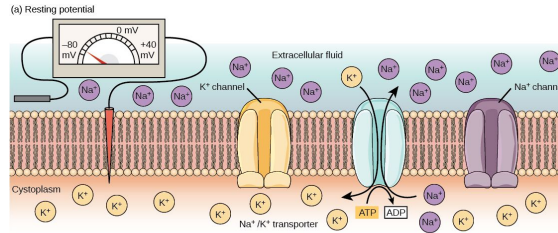


(D)

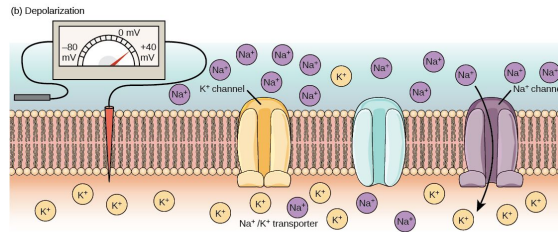


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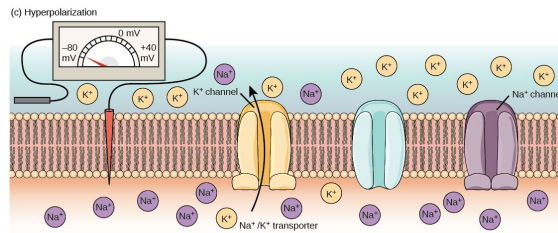
Neurons have a resting membrane potential



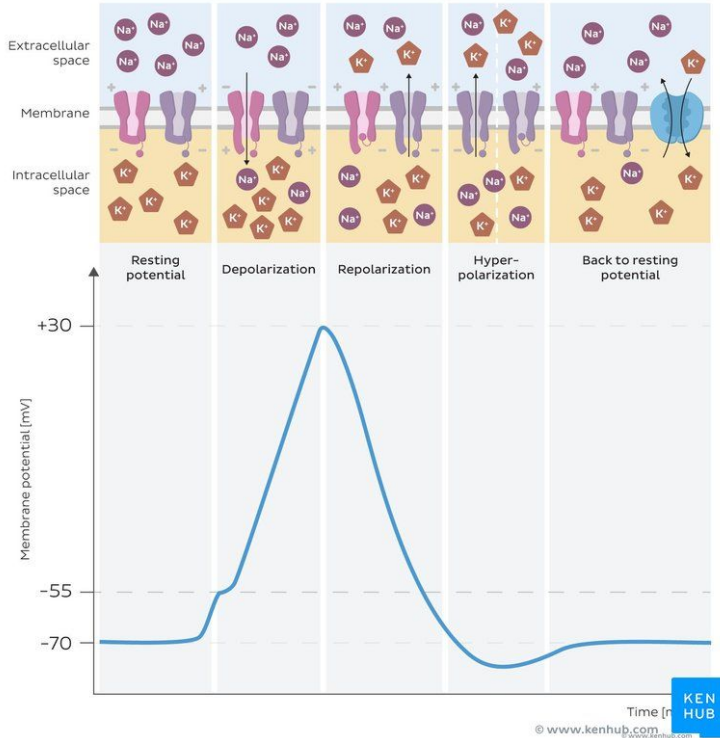
- In rest;
 - Na⁺ more abundant outside of the cell
 - K⁺ more abundant inside of the cell



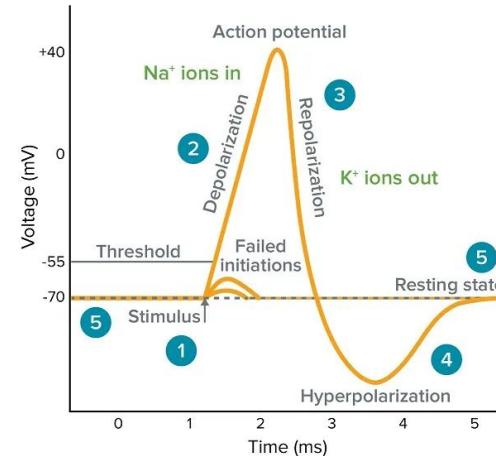
- Depolarisation;
 - Sodium channel opens to let Na⁺ into the cell
 - → Membrane potential becomes more positive
- Hyperpolarisation;
 - K⁺ channel opens to let K⁺ out of the cell
 - → Membrane potential becomes more negative



The generation of an action potential



When the membrane potential reaches a threshold: an action potential is fired.

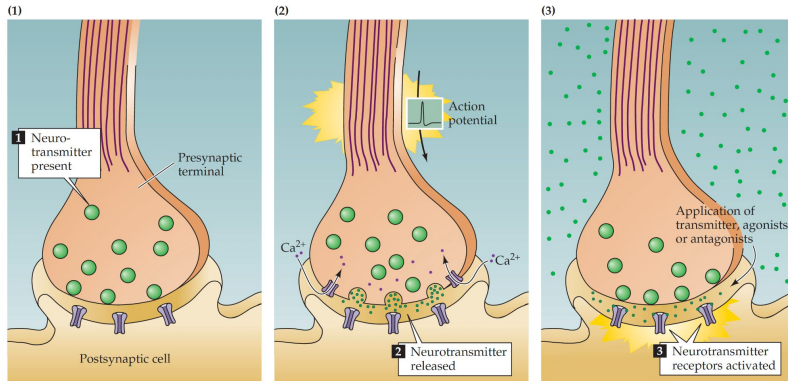


AP result in neurotransmitter release

Quite a complex and regulated process

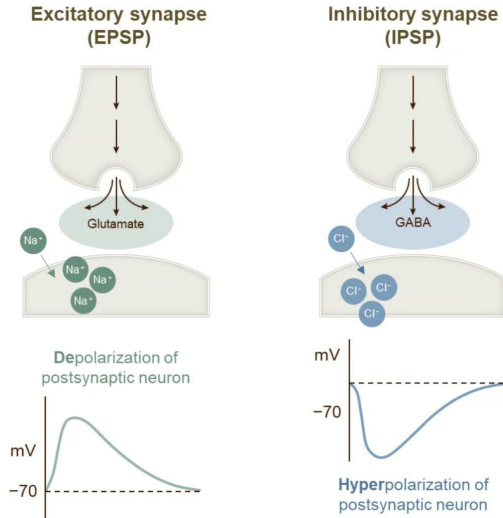
Depends on a lot of different proteins

Demonstrating the identity of a neurotransmitter at a synapse requires showing (1) its presence, (2) its release, and (3) the postsynaptic presence of specific receptors.



Different neurotransmitters results in different effects

Excitation and inhibition



Types of neurotransmitters:

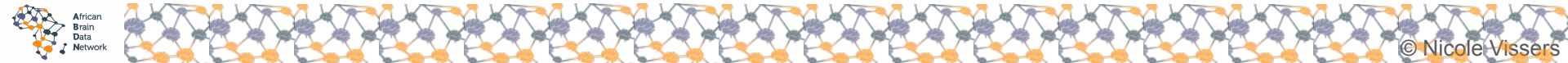
- **Glutamate:** depolarize postsynaptic neuron
- **GABA:** hyperpolarize postsynaptic neuron
- **Glycine:** hyperpolarize postsynaptic neuron

But also modulatory transmitters:

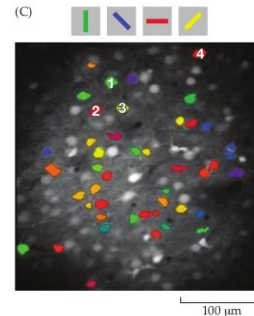
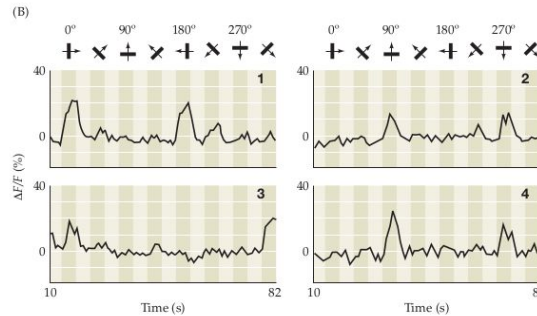
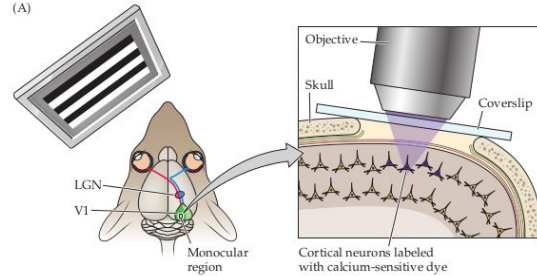
- Dopamine, Serotonin, Norepinephrine, Histamine
- Peptides
- Acetylcholine



From neurons to brain recordings



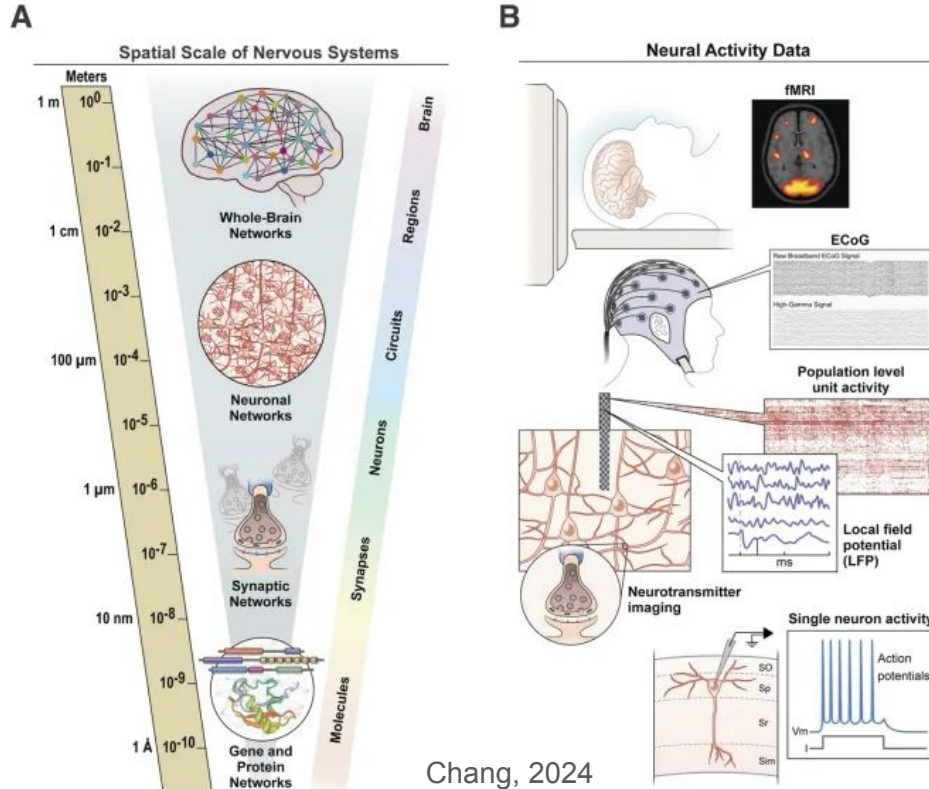
What information is contained in neural activity?



Dataset that looks at visual data:
https://observatory.brain-map.org/visualcoding/search/overview?sort_fileId=tld1_name&sort_dir=asc

Purves, 6th edition

Different scales of measurement

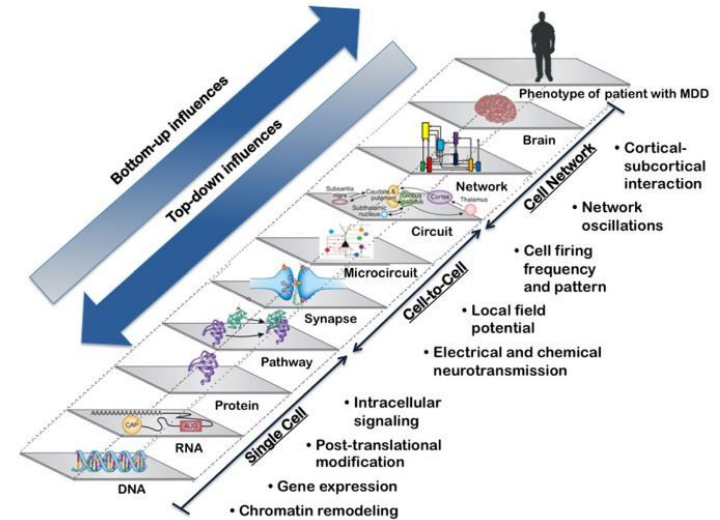


Chang, 2024

Every question we ask requires a different scientific technique.

The challenge

- A lot of different functions that cannot all be observed at the same time;
 - you get totally different picture if study a synapse vs the whole brain
- Complex; connections are not fixed and not predictable
- Everything happens at different timescales; microseconds to years
- Every method has strengths and weaknesses (focus of next lecture!)
- Everything is connected!

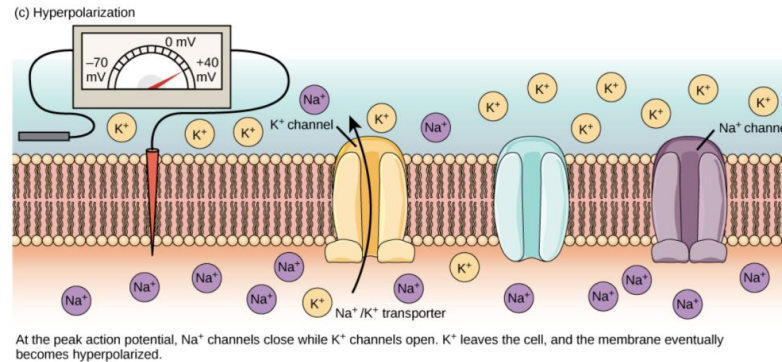


Leuchter et al., 2015

Why should we use electrophysiology?

Neurons communicate using differences in voltage;

→ Voltage differences can be measured





Open neuroscience resources



Explore the brain

Resources to access to learn more:

- Introduction to Behavioural Neuroscience

<https://openstax.org/books/introduction-behavioral-neuroscience/pages/1-introduction>

More resources can be found here:

<https://www.fens.org/news-activities/news/neuroscience-resources-available-online>

Allen Institute data:

<https://brain-map.org/our-research/circuits-behavior>



Summary

1. The brain consists of highly connected **neurons**
2. These neurons have a membrane potential that changes by **ion flow**
3. A change in ion flow results in change in **voltage** that can be measured
4. When the membrane potential reaches a threshold it fires an **action potential**
5. An action potential results in **neurotransmitter release**
6. Neurotransmitter release affects the membrane potential of other neurons
7. The **type** of neurotransmitter affects the eventual result

We can study this process at different scales and using various techniques.



What to prepare for next week!

Use github:

https://github.com/African-Brain-Data-Network/ABDN_2026_FELLOWSHIP_COHORT/tree/main/participants

Write a paragraph on:

What is a new fact you have learned about the brain?

What questions arose when learning about the brain?